



Literacy training vs. psychological inoculation? Explicating and comparing the effects of predominantly informational and predominantly motivational interventions on the processing of health statistics

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Abstract

Communicating statistics is challenging and fraught with mis-contextualization and causal misattributions. Can we train the public against statistical misrepresentations? Pre-emptive interventions against misinformation primarily include literacy tips/training and inoculation. In theory, inoculation has an additional motivational component (forewarning). However, forewarning has not been directly tested against literacy interventions, calling into question inoculation's distinction. We critique the theoretical boundary work and compare these informational and motivational interventions in the context of health statistics. The longitudinal experiment compared the effects of interventions on processing accurate and inaccurate statistics about COVID-19 vaccines and/or genetically modified organisms across digital platforms. Both interventions prevented an elevation in risk perceptions following exposure to statistical misinformation at a later time. However, literacy intervention increased risk perceptions following exposure to accurate statistics too, suggesting an additional benefit of forewarning. Those with high levels of pre-existing misinformation concern exhibited inoculation effects more strongly. We discuss the theoretical, empirical, and practical implications.

Keywords: theory-based comparison of misinformation interventions, literacy tips and training, psychological inoculation, communicating statistics (correlation vs. causation), COVID-19 vaccine and genetically modified organism (GMO) misinformation

How can we equip the public against statistical misinformation or misinterpretation of accurate statistics? Communicating statistics is challenging, and if not carefully handled, it can leave individuals susceptible to mis-contextualization of findings or misunderstanding of accurate characterizations (Peters, 2020). Misrepresentation of research findings and science-appropriating strategies also challenge the credibility of scientific evidence. Such misleading messages are hard to detect, can get news coverage due to false balance, can be more persuasive than outright inaccuracies, and often are not easily moderated on digital platforms (e.g., higher justification threshold for removal) (Kuru et al., 2021; O'Brien et al., 2021). In particular, statistical misinformation in the form of treating correlations as evidence for causal relationships is a key problem in contexts such as anti-vaccine narratives (Jamieson et al., 2024) and claims about genetically modified organisms (GMOs) (Jiang & Fang, 2019) (Supplementary Material S1). To counter such misinformation, scholars and practitioners mostly utilized post-hoc remedies such as debunking and fact-checking (Chan et al., 2017). Yet, these strategies are context-dependent and may lead to illusory truth effects (Udry & Barber, 2024). In contrast, pre-emptive interventions (a) can equip individuals with a critical lens that is generalizable and more relevant against pseudo-scientific content, (b) provide the necessary bandwidth (time and space) for discussing topics such as statistics less confrontationally, and (c) can counter misinformation when corrections are technically limited (e.g., encrypted messaging apps).

There are numerous pre-emptive interventions against misinformation. The two major interventions are literacy tips/

training and psychological inoculation. However, empirical findings offer mixed evidence for these interventions (Guess et al., 2020; Hameleers, 2023; Maertens et al., 2023). Especially in the case of literacy tips/training, there is a potential risk of inducing skepticism toward accurate information too (Guess et al., 2020). Moreover, the practical distinctions between the two interventions must be fully clarified. In theory, inoculation has an additional motivational component—forewarning—about a potential threat (e.g., misinformation) (Compton, 2021). Yet, the extent of this distinction has not been conceptually elaborated or empirically tested and not distinguished from similar educative efforts like literacy training. Also, the dosage/intensity of interventions employed in the experimental literature is more substantive in psychological inoculation (e.g., the interactive games) than in literacy intervention (e.g., bullet-point literacy tips) studies (cf. Lu et al., 2023). This ambiguity and lack of direct comparison (a) make the interpretation of empirical evidence challenging and (b) may hinder theoretical consolidation and practical insights in communication intervention science (Compton, 2020). This is problematic at a time of extensive efforts in comparing interventions without a critical assessment of their underlying similarities and differences. At large, this also runs the risk of unproductive boundary work and competition that may fuel disciplinary silos.

Against this background, the current study critically compares the research on literacy training and inoculation interventions. We then propose and test three strategies to rigorously compare the two interventions: (a) holding

constant the underlying informational (knowledge and skills) component, (b) testing the absence and presence of motivational (forewarning) component, (c) testing the effects of a set of conditioning factors on the underlying informational (with *news literacy behaviors*) and motivational (with *misinformation concern* and *message format*) components to further unpack how the two components might operate independently. These three conditioning factors may shape the effectiveness of the two interventions' underlying informational and motivational components. A longitudinal experiment was fielded to compare the interventions' effects on individuals' processing of both accurate and inaccurate statistics about the coronavirus disease 2019 (COVID-19) vaccines and, as a partial conceptual replication, about GMOs. This study provides the first direct and theory-driven testing of the major pre-emptive misinformation interventions, thereby offering some explanation for (a) the inconsistent or unexpected empirical results, (b) stronger effects observed in inoculation relative to literacy training, and (c) the underlying theoretical similarities and distinctions between the interventions. Next, we first review research on literacy training and inoculation, detail our empirical strategy and the context, and finally explain the design and hypotheses.

Literacy tips/training interventions

Widely considered a traditional educative approach, literacy training equips individuals with knowledge and skills to be more attentive and critical in subsequent information processing in various contexts (Amazeen et al., 2022; Jones-Jang et al., 2021). Digital media and news literacy interventions, such as providing tips on how to spot fake news, can be effective against misinformation and boost truth discernment (Guess et al., 2020). For example, news literacy tweets, encouraging people to be more critical, were able to change misinformed beliefs about GMOs and flu vaccines (Tully et al., 2020), and train individuals about reverse-searching out-of-context images (Qian et al., 2022).

However, not all types of literacy training are effective, and some can even be counterproductive. For example, video-based interventions about misinformation (e.g., three tips to identify fake news) did not change the ability to recognize correct or fake news (Ali & Qazi, 2023). News literacy tips can reduce belief in real headlines too (Guess et al., 2020). Other literacy interventions generated cynicism and failed to increase the effectiveness of corrections (Vraga et al., 2021a). Also, reduced trust in response to media literacy tips spilled over to the assessments of accurate content (Hoes et al., 2024). These inconsistent findings present a contested picture of the usefulness of literacy interventions and raise questions about the mechanisms underlying these effects that the literature still needs to address. These findings also stand in contrast to those of a competing line of work: psychological inoculation.

Psychological inoculation interventions

Outperforming literacy training in scholarly popularity and the consistency of its effectiveness is psychological inoculation. Based on theorization that utilizes the vaccine analogy (Compton, 2020), inoculation has regained substantive attention over the last decade, fueling many studies, reporting of notable effects, and increasing implementation by practitioners,

governments, and digital media platforms. Inoculation refers to building defensive preparation against future persuasion attempts by providing information about tactics (pre-bunking) and forewarning about a threat (e.g., of a persuasion attempt or misinformation exposure), similar to the mechanisms of medical vaccination (Lewandowsky & van der Linden, 2021). Recent studies exhibit multi-component structure and intense dosage in treatments (e.g., interactive games). This resulted in an expanding repertoire of different types of inoculation, such as (a) passive vs. active inoculation, (b) specific vs. technique-based inoculation (Roozenbeek, Traber et al., 2022), and (c) inoculation across different modalities (textual visual, video-based, etc.). However, we can argue that all of these versions share the two fundamental underlying components: information/skill and forewarning as a motivational trigger (cf. Roozenbeek, van der Linden et al., 2022).

Extensive work shows the effectiveness of this strategy, seemingly with only a few caveats in the results. Passive inoculation is traditional information provision while active inoculation refers to more engaging activities such as playing gamified activities, which may be more effective (Basol et al., 2020). Technique-based inoculation, which explains flawed arguments such as false balance media coverage, effectively lowered misperceptions about climate change (Cook et al., 2017). Logic-based pre-bunks (e.g., oversimplification tactic) also reduced climate change misinformation (Vraga et al., 2019). Individuals can be inoculated against a variety of misinformation tactics (e.g., emotional manipulation, false dichotomies) (Roozenbeek, van der Linden, et al., 2022). A limitation is that inoculation effects are mostly observable in cross-sectional studies where people are tested immediately after training, making their practical and social significance limited. The decay of effects is extensive if people are not "re-inoculated" across time (Maertens et al., 2023).

Literacy training or inoculation?

How do literacy training and inoculation differ from each other? Notably, many studies approached literacy training and inoculation either (a) interchangeably or (b) distinctly without clarifying how they were distinct. Indeed, he studies either did not distinguish between the two interventions at all or were vague about their distinctions. For example, "a media literacy intervention that warned people about mis- and disinformation" found unintended effects such as a reduction in believing accurate claims (Hameleers, 2023). It is not clear whether it is literacy or inoculation when the literacy intervention consists of a warning against misinformation. Another study examined the effects of interventions including "news media literacy" on the perceived accuracy of real news; yet, this intervention was about the provision of informational statements about the level of misinformation in the society, which is similar to the forewarning in inoculation (van der Meer et al., 2023). A recent study included "accuracy prompts" into inoculations too (Pennycook et al., 2023). Yet, accuracy prompts can be considered a form of motivational trigger, like forewarning. Implicit in an accuracy appeal is the *threat* of inaccuracy. Also, being accurate is a fundamental purpose of literacy training as well. Finally, synthesis research further highlights the vagueness of the literature: A recent systematic review on media literacy interventions (Droog et al., 2024) included inoculation studies too. On the other hand, a meta-analysis of inoculation effects

included studies (e.g., [Vraga et al., 2019](#)) that treated post-hoc or simultaneous corrections as a type of “therapeutic inoculation” ([Lu et al., 2023](#)). Despite providing important findings, prior work did not pay enough attention to these interventions’ underlying similarities and distinctions. We argue that there is a lack of attention to and conflation of the interventions’ predominantly informational and predominantly motivational components.

Furthermore, direct critiques on inoculation call into question its coherence and thus cast doubt on its distinction from literacy interventions. First, some scholars argue that inoculation is theoretically incoherent, particularly in its characterization of misinformation as a virus ([Williams, 2023](#)). Others point out that inoculation is too specific and not applicable to a “diverse array of real-world news content” ([Guess et al., 2020](#)), although this critique can be viewed as less relevant to technique-based inoculations as they provide more generalized and procedural knowledge ([Roozenbeek, Traber, et al., 2022](#)). Second, while some scholars acknowledge that inoculation in real life can never be pre-emptive, they also claim that this does not matter from a practical perspective ([Basol et al., 2020](#); [Lewandowsky & van der Linden, 2021](#)). However, this temporal fuzziness is still problematic, leading some to question the central promise of the vaccine analogy and argue that inoculation could alternatively be characterized as the processing of “(two-sided) sequential messages” ([Wood, 2007](#)). Similarly, the threat component cannot be easily isolated in real-life settings, which may leave inoculation at least as a method, if not the theory, hard to distinguish from literacy interventions. As mentioned, some recent research conceptualizes post-hoc or simultaneous corrections as “therapeutic inoculation” ([Vraga et al., 2019](#)). Whether post-hoc strategies can be considered inoculation is yet an unsettled debate ([Compton, 2020](#)), and it is a debate that can further erode the distinctions between inoculation and literacy training. Finally, other notable critiques of inoculation, such as its deterministic outlook (misinformation virus, vaccination analogy) and ethical issues, such as its individualization of the problem—misinformation as the disease and not the symptom of larger societal problems—can exasperate the theoretical coherence issues both directly and indirectly ([Freiling et al., 2023](#); [Scheufele, 2022](#); [Williams, 2023](#)). While these critiques might also be relevant for various other interventions against misinformation, inoculation is arguably the most deserving target.

Against this background, we argue that it is crucial to compare these two interventions directly with a focus on their underlying similarities and differences. Why is such a comparison important? For one, as reviewed above, the effects of inoculation interventions can be characterized as more robust and consistent than those of literacy training interventions. Second, literacy interventions backfire (relatively) more frequently in various contexts. Yet, it is unclear if and to what extent these empirical trends are primarily attributable to their theoretical differences (e.g., forewarning) or merely to the differing levels of intensity/dosage of the interventions in the design of the studies. Third, the vagueness and the conflation with which the literature treated these interventions might at least partly be indicative of theoretical boundary work. This then creates challenges for the accumulation of empirical and practical insights in communication intervention science (cf. [Compton, 2020](#)).

Three empirical strategies to explicate and compare the underlying components

We offer three empirical strategies that can be exploited to compare the two interventions: (a) keeping the strength/dosage and the knowledge/skills content the same (informational component), (b) testing the absence/presence of the forewarning component (motivational component), and (c) testing a set of key conditioning factors to understand how underlying informational and motivational part might operate independently.

Holding constant the strength/dosage of interventions

The informational component of both interventions should be kept constant and have equal intensity for an equivalent comparison. Most literacy interventions against misinformation were tested for their effectiveness via experiments that included brief and very short treatments, such as providing news literacy tips with bullet points ([Kozyreva et al., 2023](#)). Some interventions consisted of messages such as “1) check sources of messages, 2) look for factual information and 3) look for logical argumentation styles” ([Hameleers, 2023](#)). These could be considered small-scale and general statements. Exceptions exist; a few studies had more substantial dosages in the form of more extensive training ([Moore & Hancock, 2022](#)) and targeted more specific skills ([Qian et al., 2022](#)). Still, collectively, literacy studies have lower dosages than inoculation studies ([Basol et al., 2020](#)). We argue that more substantive literacy training can be developed to include explanations of logic (like technique-based inoculations), both accurate and inaccurate examples to foster learning (like forewarning in inoculation), and practice quizzes (like gamified and active inoculations) to make them more engaging and on par with the intensity of inoculation treatments widely deployed in the literature. Both interventions’ informational components can include definitions, explanations, examples, quiz questions and/or gamification (cf [Basol et al., 2020](#)). If we include these diverse informational components in both interventions, they will be equivalent in intensity and, hence, directly comparable. This stands in contrast to prior studies that compared numerous misinformation interventions by assuming those interventions as being categorically different ([Arechar et al., 2023](#); [Fazio et al., 2024](#); [Hoes et al., 2024](#); [Spampatti et al., 2024](#)).

Varying the forewarning component

The presence of forewarning should be directly tested. Forewarning, as a key component in inoculation ([Compton, 2021](#)), involves informing individuals that they would be exposed to mis/disinformation ([Cook et al., 2017](#); [Maertens et al., 2023](#)). This constitutes a motivational triggering that could be considered more distinct from the informational component focusing on knowledge and skills ([Banas & Richards, 2017](#)). This motivational trigger acts as the driving force for individuals to be more specific when they mobilize their attention and critical information processing. Yet, literacy interventions might have (implicit) threat components too (e.g., [Hameleers, 2023](#); [van der Meer et al., 2023](#)). In practice, it is hard to imagine scenarios for literacy interventions where the target does not refer to a threat, at least indirectly. Threats in literacy intervention could be misinformation or misunderstanding of an issue, and this may be done either explicitly or implicitly. Also, exposing individuals to a *small dose of misinformation* (a component in

inoculation) is practically the same as providing *examples of problematic content* in literacy training. Literacy approaches provide examples to aid comprehension (e.g., providing incorrect cases to identify). This makes it imperative to test the absence and presence of a forewarning component *explicitly and boldly by characterizing literacy training as predominantly informational and inoculation as predominantly motivational*. Forewarning should make inoculation a more effective intervention due to its mobilization of critical mindset.

Testing the conditioning factors on the underlying predominantly informational and predominantly motivational components

Third, to further evaluate whether and how informational and motivational components might operate independently, the conditioning roles of a set of factors that target these components could be investigated. This will reveal the relative importance of these components by unveiling potential contingent effects (cf. Holbert and Park, 2019). In this study, we specifically test the influence of three conditioning factors—two of them are pre-existing individual differences and the third one is a contextual and manipulated factor: (a) individuals' pre-existing news literacy behaviors, (b) misinformation concerns, and finally, (c) the format (digital platform) in which evaluated messages appear. The factor of news literacy behaviors taps the underlying informational (facts and explanations) component in both interventions while the factors of misinformation concern and the message format tap the motivational (forewarning) component in inoculation (Supplementary Material S1).

News literacy behaviors (NLBs), such as cross-checking the source or paying attention to the numerical information in news, could shape how individuals process statistical evidence (Vraga et al., 2021b). NLB could change whether and how individuals may act on their methodological knowledge (e.g., correlation is not causation) by tapping the informational component. If people engage in NLB more (e.g., item "examining numerical evidence in detail"), they are more likely to act in response to the interventions. High NLB individuals would pay more attention to numbers and details to being with.

Misinformation concern is introduced as a motivational conditioning factor concerning the forewarning component (cf. Altay & Acerbi, 2023). Misinformation concern is independent of the prevalence of misinformation, showing its potential to condition key outcomes such as misinformation mitigation (Knuuttila et al., 2022). Specifically, misinformation concern could condition how much the forewarning component may trigger critical vigilance. Those with higher levels of pre-existing misinformation concern should be more receptive to threat messaging in forewarning because these individuals would have a lower threshold of reacting to threats. With higher baseline vigilance, individuals in inoculation intervention should display critical information processing more strongly and speedily or sustain their hawkish vigilance for a longer time.

Message format (news vs. social media post) can tap forewarning uniquely. The format of messages that individuals process following the completion of interventions may condition the effects of the interventions. Platforms provide us cues regarding the credibility of the content they host (e.g., the reputation heuristic in source credibility, Metzger & Flanagin, 2013). Platforms such as news media and social media differ regarding the probability of being exposed to

misinformation; prior work unsurprisingly shows that the information delivered on social media is deemed less credible than that delivered as news (Wasike, 2017). Hence, individuals should be more critical of information on social media as (a) social media is known to be riskier (i.e., the reputational heuristic) and (b) literacy and inoculation messages have been warning individuals, for some time, to be especially vigilant on social media (cf. Lu et al., 2023). Thus, the conditioning factor of message format may help us test the sensitivity of forewarning, as people might assume a more critical lens while encountering a social media post. Given the centrality of forewarning, we thus focused on not only a pre-existing individual difference factor (misinformation concern) but also on this manipulated contextual moderator (message format) to unpack how forewarning operates.

Empirical context of health statistics: correlation vs. causation

The current study offers an empirical test in the context of misleading health statistics, an important area that has been overlooked despite its centrality in health misinformation (cf. Peters, 2020). The use of scientific-sounding information is one of the key tactics employed in pseudo-scientific misinformation (Caulfield et al., 2019; O'Brien et al., 2021); such misinformation can reduce vaccine intentions more strongly (Loomba et al., 2021). Various reports on COVID-19 vaccines misrepresent studies by making unwarranted causal claims and exaggerating the prevalence of serious side effects (Supplementary Material S1). For example, such messages focus on excess deaths and vaccination rates across countries by cherry-picking data (Reuters Fact Check, 2022). Data from the Vaccine Adverse Event Reporting System (VAERS) in the U.S. is misrepresented by anti-vaccine individuals on social media; and this dataset confuses a majority of the public (Jamieson et al., 2024). Similarly, rumors about genetically modified soybeans causing cancer on Weibo included unsubstantiated "scientific evidence" in support of claims (Jiang & Fang, 2019). A content analysis of the GMO soybean—cancer link scare in China found that people paid attention to the conclusion headlines but did not pay much attention to how this conclusion (GMO soybean—cancer link) was reached (Jiang & Fang, 2019). Such statistical misinformation, most commonly in the form of causal misattributions, gains prominence at a time of growing culture of "I do my own research," (Chinn & Hasell, 2023) as individuals in such movements claim correlations as causal evidence on blogs such as *Substack* or hard-to-monitor messaging applications like WhatsApp and Telegram (Supplementary Material S1). As the non-expert public may be more susceptible to logical pitfalls in causal spins and exaggerations (Adams et al., 2017), it is important to tackle statistical misinformation, and more specifically, the common mistake (or tactic) of drawing unsubstantiated causal inferences such as vaccines *causing* death or GMO *causing* cancer.

To our knowledge, no research tested the effects of pre-emptive interventions on misleading use of statistics. Educational research shows that students who attended an introductory course on causality improved their ability to correctly identify causality in news headlines (Mueller & Coon, 2013). While the larger public cannot take an introductory course on causality, this suggests that comprehensive training interventions in the form of an explanatory, interactive, concise session can be promising. Yet, only a few studies

target statistical misinformation and only do so in the form of very specific post-hoc interventions. One study examined the effectiveness of debunking misleading causal claims; however, these corrected claims were anecdotal evidence (Vraga et al., 2019). Another study provided corrections against misleading news headlines by providing statements on causality (Irving et al., 2022). While these corrections offer promising refutations, it is unclear what derives the correction effects since correction messages often include elements directly refuting specific misinformation. Second, it is not clear if all individuals would understand statements like “correlation does not imply causation” unless they are familiar with concepts like “correlation” and “causation.” In conclusion, misleading health statistics offer an ideal empirical context to compare pre-emptive interventions in a more realistic- and scientific-looking misinformation space. It is a domain where the cognitive dimension of science literacy can be targeted in a specific way (Howell & Brossard, 2021) by informing and motivating individuals for being critical of statistical claims.

The current study

We implemented a two-wave longitudinal survey experiment in Singapore in late 2023. In the Time 1 survey, respondents were allocated to one of the three conditions: (a) control task condition (no intervention), (b) literacy training (intervention), and (c) inoculation (intervention). The interventions were online training modules with multiple stages (Figure 1, top panel). In the Time 2 survey, respondents were allocated to one of the five conditions: (a) inaccurate vaccine information in news story format, (b) inaccurate vaccine information as social media post, (c) accurate vaccine information in news story format, (d) accurate vaccine information as social media post, and (e) inaccurate GMO information in news story format. Thus, each respondent was assigned to one of the 15 groups (Figure 1, bottom panel).

The first outcome variable is the perceived credibility of messages about vaccines and GMO safety. The goals of pre-emptive interventions are to reduce and/or prevent individuals' likelihood of conferring credibility to inaccurate claims (i.e., “*vaccines are dangerous as they lead to heart attacks/ GMO consumption cause cancer,*” as shown by statistical evidence) and/or to increase or preserve the credibility of accurate claims (i.e., “*vaccines are safe and serious side effects are rare,*” as shown by statistical evidence). Simultaneously, these differential credibility assessments are expected to be accompanied by corresponding shifts in risk perceptions and intentions. Risk perceptions constitute an important outcome in determining if belief updating occurs, a more substantive outcome than the more situational message credibility assessments (Ferrer & Klein, 2015). Similarly, intentions of getting vaccinated (if health authorities recommend) and not showing specific avoidance towards GMOs (due to inaccurate beliefs), are important behavior-relevant and future-oriented outcomes. Behavioral intentions can offer more robust, tangible, and conservative evidence for the practicality of findings (i.e., being most proximal to the behavior of getting vaccinated) (Ajzen & Manstead, 2007). These intentions show the extent to which misinformed beliefs about issues such as COVID-19 vaccines and GMOs may hinder public health efforts in infectious disease control or cause unnecessary public health scares, undermining global food supply dynamics or diverting public attention away from more pressing issues

(e.g., more equitable distribution of environmental resources, to fight world hunger) (Oliver, 2014).

Thus, the following hypotheses are proposed: Compared to control, both literacy training and inoculation messages will lead to (a) increased discrediting of misinformation messages (H1a), (b) decreased risk perceptions about COVID-19 vaccine and GMOs (H1b), and (c) increased intentions to get vaccinated or not specifically avoiding GMOs consumption (H1c) while there will be no difference between interventions in the processing of accurate statistics. We expect both interventions to have this effect because both include training about correlation vs. causation. Further, compared to literacy training, inoculation messages will exhibit stronger effects observed in H1, because inoculation has the additional component of forewarning which should trigger and mobilize a critical lens on statistical claims more strongly (H2). We expect that those with greater news media literacy will be more responsive to both interventions as both include the informational component (H3) while those with greater pre-existing misinformation concern will exhibit stronger effects (as described in H1) for inoculation (H4). We also expect that the effects of inoculation will be stronger for misinformation presented in social media posts than for news reports (H5) because individuals will act more strongly on forewarning during the processing of content encountered on social media—the platform with greater (real and/or perceived) risk for hosting misinformation.

Methods

Dataset

We received ethics approval at the Institutional Review Board of the National University of Singapore. The protocol and hypotheses are preregistered: <https://aspredicted.org/ZG4_57Q> and the data and code are public <https://osf.io/gmwra/?view_only=188dc3411fba4f1db6676600db0f67bd>. Some parts of preregistration were not as specific and detailed as articulated in the manuscript; these are clarified in [Supplementary Material S2](#) as deviations. We collected data through a two-wave longitudinal online panel survey experiment on a non-probability sample of individuals who are legally adults in Singapore (21 and older) as obtained through *Qualtricssm* in September 2023. Quotas were used to maximize the demographic representations. The sample size for Wave-1 (W1) was $N = 2,061$, and for Wave-2 (W2, recontact) was $N = 1,278$, however, during data collection, duplicate cases were detected and removed, which amounted to 77 cases across the waves (14 in W1, 63 in W2). W1 valid sample size is hence 1984. All withdrawal requests following debriefing are honored ($N = 22$), hence W1 effective (final) sample size is 1962. After debriefing withdrawal requests, and removal of speeders and laggards based on pre-set criteria (see [Supplementary Material S2](#)), the effective sample size for the main (longitudinal) analysis (hypothesis testing on respondents for whom we have valid two-wave data) is $N = 1104$. Comparing W1 valid $N = 1962$ to W2 valid $N = 1185$ before removing poor quality respondents (e.g., speeders), the response rate across waves is 60%. Descriptive details for the recontact sample are in [Supplementary Material S2](#) and the attrition analysis in [Supplementary Material S3](#) provides details for each of the W1 and W2 samples separately too.

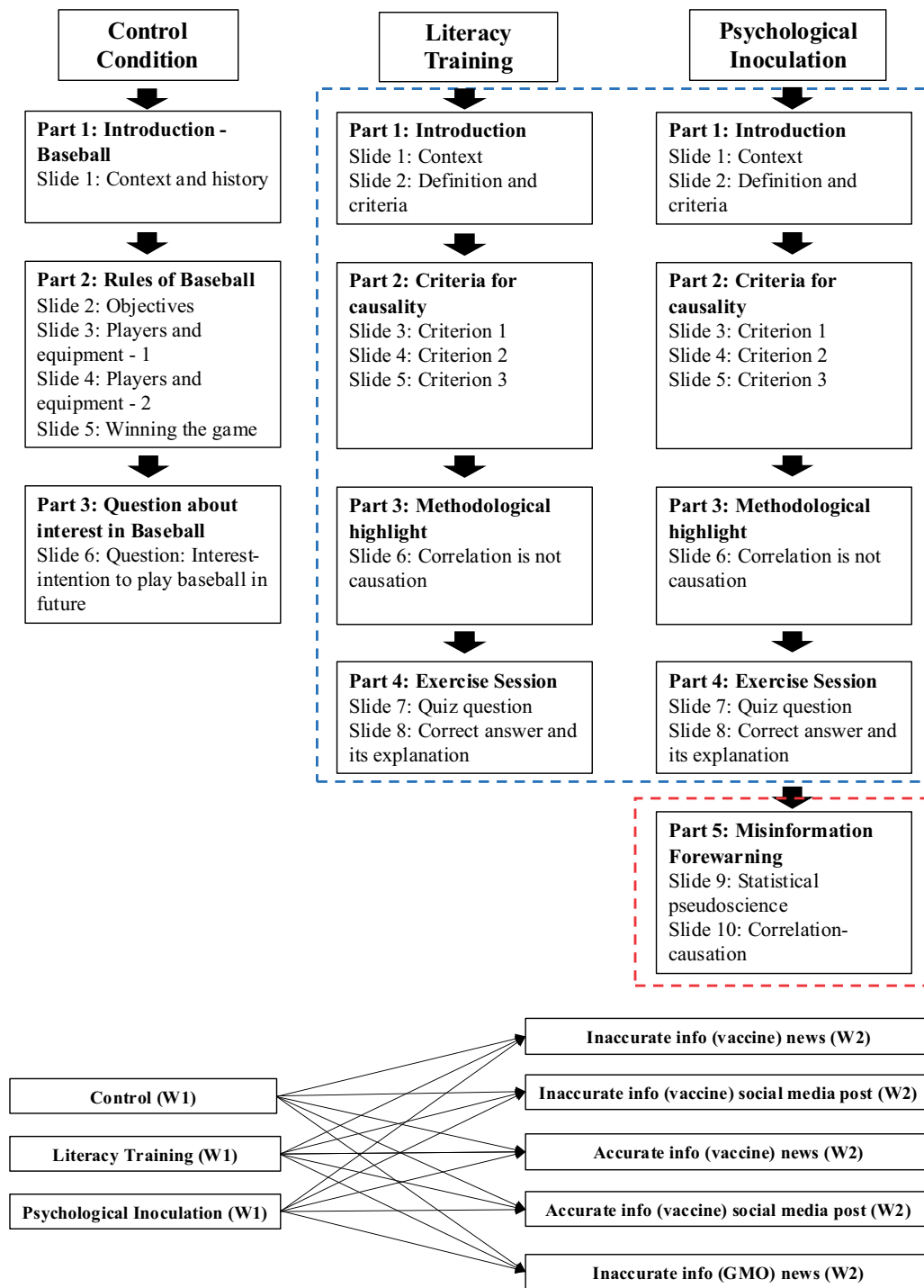


Figure 1. Structure of Wave-1 (W1) control and intervention modules (upper panel) and W1 by W2 conditions across the waves (lower panel).

Procedure

At W1, participants read the consent form and answered an attention check question. Those who failed the attention check were not allowed to continue. After pre-tests, control measures, and unrelated buffer questions, respondents were randomly assigned to one of the three conditions, followed by manipulation checks. The average number of days between W1 and W2 was 18.9 (median = 21 days). The W2 survey was shorter and exposed the participants randomly to

one of the five manipulation messages and then outcome measures. It additionally included a set of interference measures to account for information exposure between the waves and the full debriefing at the end.

Manipulations

The conditions emerge from the crossing of manipulations at both waves. W1 has three groups ($C1_{W1}$ = control, $C2_{W1}$ = literacy training, $C3_{W1}$ = inoculation) while W2 has 5 groups

(C1_{W2}= vaccine misinformation news, C2_{W2}= vaccine misinformation post, C3_{W2}= vaccine accurate information news, C4_{W2}= vaccine accurate information post, C5_{W2}= GMO misinformation news), resulting in a total of $3 \times 5 = 15$ conditions, as seen in Figure 1. GMO misinformation condition provides a partial conceptual replication. Literacy intervention of training about correlation vs. causation has four components as seen in Figure 1: (a) context and definitions, (b) explanations for criteria of causality, (c) methodological highlight, (d) practice question and correct answer explanation. Inoculation intervention has these same components, and additionally, forewarning. W2 messages were adopted from real misinformation examples (for misinformation conditions) and published scientific research findings (for accurate information conditions) (Supplementary Material S4). To make the conditions comparable, exact statistics from a published article on vaccine safety were used in both misinformation and accurate information conditions. These same accurate statistics are used misleadingly in the misinformation conditions, similar to the real-life misinformation examples we collected (Supplementary Material S4). The misinformation stories point out presumed “evidence” for vaccines *causing* very frequently observed serious side effects, such as those leading to deaths (and the same for GMOs *causing* allergy, cancer, and death). However, GMO conceptual replication has no accurate information condition due to ecological validity and power/sample size concerns. See Supplementary Material S4 for reasoning details and citations for these decisions. Supplementary Material S2 provides power analysis. Manipulation checks showed that manipulations worked (Supplementary Material S5), balance tests documented the equivalence of conditions (Supplementary Material S6), and cell Ns were balanced as well (Supplementary Material S7).

Measures are described below, and descriptive statistics are provided for the longitudinal data. Descriptive results for W1 and W2 are provided separately in Supplementary Material S3. Measures were adopted from prior research, and further details are provided in Supplementary Material S2.

Measures

Independent variables (manipulation factors)

Intervention, as a factor, has three groups (control, literacy training, and inoculation). Accurate vs. inaccurate information, as a factor, contrasts accurate (C2, C3, coded 0) and inaccurate (C1, C2, C5, coded 1) messages in W2. News vs. social media, as a factor, contrasts news story format (C1, C3, C5, coded 0) with social media format (C2, C4, coded 1).

Outcome measure 1—perceived credibility of messages (post-test at W2)

The three items were adopted from extensive recent work on credibility (Albarracín et al., 2024) “How [informative/reliable/accurate] do you find the information you just read?” with five response options ranging from “Not at all [informative/reliable/accurate]” to “Extremely [informative/reliable/accurate]” ($\alpha = 0.90$). They were averaged into an index where greater scores represent higher levels of perceived credibility ($M = 0.44$, $SD = 0.23$).

Outcome measure 2—risk perceptions (pre- and post-test at W1 and W2)

Risk perceptions for both COVID-19 vaccines and GMO food were assessed by three identical questions each (Albarracín

et al., 2024): “[How risky do you think/How concerned are you about/How worried are you about] the side effects of COVID-19 vaccines [are]?” with 5 options ranging from “Not at all risky” (1) to “Extremely risky” (5). For GMOs, all wording is the same, and “COVID-19 vaccines” is replaced by “eating genetically-modified foods.” The three items had good reliability ($\alpha = 0.92$ for vaccine and 0.92 for GMO) and were averaged into an index that ran from 0 to 1 ($M = 0.44$, $SD = 0.25$ for vaccine, $M = 0.53$, $SD = 0.26$ for GMO). We also created a difference score by subtracting W1 from W2 scores, where positive values correspond to increased risk perception over time and vice versa for negative values. The differences are then averaged into an index ($\alpha = 0.77$, $M = 0.0$, $SD = 0.19$ for vaccine risk difference score index and $\alpha = 0.86$, $M = 0.06$, $SD = 0.23$ for GMO).

Outcome measure 3—behavioral intentions (pre- and post-test at W1 and W2)

The vaccination intention item asked, “How likely are you to get a COVID-19 vaccine (or booster shot) in the future if health authorities recommend it for your age group?” with five options from “Not at all likely” (1) to “Extremely likely” (5). ($M = 0.52$, $SD = 0.31$). The GMO intention item asked, “How likely are you to avoid buying food that is genetically-modified (GMO) in future?” with 5 responses ranging from “Not at all likely to avoid” to “Extremely likely to avoid”. GMO item was then reverse coded to make directionality the same as the vaccine measure, with greater values showing less intentional avoidance ($M = 0.42$, $SD = 0.30$). We also created difference scores for these single item intentions ($M = 0.01$, $SD = 0.22$ for vaccine and $M = -0.01$, $SD = 0.14$ for not avoiding GMO).

News literacy behaviors (pre-test at W1)

Nine items measure behaviors that respondents engage in while reading news and information (Wu et al., 2023): “When you are reading digital news articles on the Internet, how often do you ...” with items such as “Examine the numerical and statistical information in detail,” (Supplementary Material S2) and with five response options ranging from “Never” to “Very frequently.” The nine items had good reliability ($\alpha = 0.92$) and were averaged into an index that ran from 0 to 1 ($M = 0.52$, $SD = 0.20$).

Misinformation concern (pre-test and post-test at W1 and 1 item in post-test at W2)

Three items were adopted from recent work (Altay & Acerbi, 2023), such as “How concerned are you that you might be exposed to false or misleading information?” Reliabilities were high ($\alpha = 0.83$ at *pre-test W1*; $\alpha = 0.85$ at *post-test W1*), and items were averaged into an index ($M = 0.67$, $SD = 0.21$). Three difference score indexes/measures were computed (W1 *post*—W1 *pre-test difference* index, W2 *post*—W1 *pre-test difference* index, W2 *post*—W1 *post-test difference* score (not an index, it is only one item in W2). The reliability score is not applicable because the difference scores compare pre- and post-manipulations across different interventions. Pre-test misinformation concern is used as a moderator, while the post-test measure is part of the manipulation checks (Supplementary Material S5).

Analytical strategy

Both cross-sectional and longitudinal analyses are conducted with ANOVA and OLS regressions (Supplementary Materials S8 and S9). The primary analyses were based on the two-way ANOVA formula, " $SS_{total} = SS_{intervention} + SS_{accurate\ vs.\ inaccurate} + SS_{intervention \times SS_{accurate\ vs.\ inaccurate}} + SS_{residuals}$ " while moderation hypotheses had an additional interaction term. Assumption checks for ANOVA were largely met too (Supplementary Material S2). Multiple (alternative) models are tested for each hypothesis, including among all conditions and smaller subsets of conditions; all results are presented either in the main report or in Supplementary Materials. This allows us to test the different components in W2 manipulations involving accurate vs. inaccurate information (and conceptual replication with GMO) both independently and in a combined way. This strategy also helps avoid the testing of very complicated three-way or four-way interactions. As the equivalence of conditions was achieved in pre-test and demographic measures, the main models do not include additional controls, but we also tested models by controlling for demographic and control variables; we measured age, female, Malay, Indian, and Other race categories (Chinese reference), income, and education levels, ideology, news exposure, and trust in science (see Supplementary Materials S2 and S10).

Results

Overall, perceived credibility evaluation results showed no across-the-board intervention effects. However, among respondents with high misinformation concern levels, inoculation intervention had a significant impact. There were also significant intervention effects on risk perceptions and behavioral intentions. Finally, the message format did not moderate the intervention effects, although it had notable main effects and interaction with the accuracy vs. inaccuracy of messages.

Credibility evaluations (H1a, H2a)

Credibility of messages did not differ significantly between the W1 conditions in any combination of W2 conditions; that is, interventions did not work across the board for this outcome (Table 1, Column 1, Rows 1–3; and Supplementary Material S9). Participants assessed misinformation as less credible than accurate information ($M_{diff} = 0.10, p < 0.001$, Tukey's test); this main effect persisted during interaction with intervention manipulation too (Column 1, Row 2), but it was not moderated by interventions (Column 1, Row 3; see Supplementary Material S8 for visualization of these null effects). Thus, neither H1a (control vs. interventions) nor H2a (literacy vs. inoculation) are supported.

Risk perceptions (H1b, H2b)

Risk perceptions differed significantly across conditions in notable ways. Among all conditions, there was a significant interaction between W1 intervention and W2 accuracy vs. inaccuracy of message content as seen in Table 1 (Column 2, Row 3). Such that, among the W1 control condition, those who saw misinformation at W2 had significantly higher risk perceptions at W2 about COVID-19 vaccines and GMOs than those who saw accurate information (Column 2, Row 4). However, for those in either of the W1 intervention conditions (literacy training and inoculation), there was no such difference. The same pattern was observed among all vaccine

conditions only (excluding GMO) as well (Supplementary Material S9). These results extend to the within-person longitudinal change as well, where the outcome variable is the "W2 post-test—W1 pre-test" difference score in risk perceptions (Column 3, Rows 3 and 4). Both Figure 2A (cross-sectional) and Figure 2B show notable negative influence of misinformation (in gray) in the absence of interventions (lowest two distributions). There is partial support for H1b and H2b.

Also, there was an unexpected result among accurate vaccine information conditions; there was a main effect of interventions for longitudinal risk perception (Column 3, Row 6). Literacy intervention significantly increased risk perception compared to control (Column 3, Row 7). As seen in Figure 2C; literacy intervention (in dark gray) increased risk perceptions in response to accurate (vaccine safety) statistics, compared to control (in blue). While the omnibus F test was not significant ($p < 0.10$), the same difference (Tukey's post-hoc) was significant for the cross-sectional risk perception, too (Column 2, Row 6).

Behavioral intentions (H1c, H2c)

Interventions shaped intentions too, although in smaller and less consistent ways than risk perceptions. First, there was an overall main effect of W1 intervention for cross-sectional intentions. Such that, those in inoculation intervention, compared to control, displayed higher intentions overall following exposure to both accurate and inaccurate information ($M_{diff} = 0.05, p < .05$; main effect in Table 1, Column 4, Row 1). Figure 2D illustrates this, where inoculation (in light gray) displayed a more negatively skewed distribution. While there was no significant interaction with accurate vs. inaccurate messages (Column 4, Row 3), post-hoc tests revealed that the inoculation group that saw accurate information had substantively higher intentions (a) than the control condition that saw misinformation (Column 4, Row 4) and (b) than those in literacy training who saw inaccurate information, although this difference between the two interventions was not significant (Column 4, Row 5, $p < 0.10$). In Figure 2E, the inoculation group exposed to correct information (second distribution from the top, in blue) exhibited a strongly negative skew as compared to the control group exposed to misinformation (second group from the bottom, in gray). These results did not hold when analyzed only among vaccine or GMO conditions. Among accurate vaccine conditions, inoculation seemed to lead to higher intentions at W2 than literacy training. However, this was not significant as well (Column 4, row 7, $p < 0.10$). Results were similar for the longitudinal difference scores for intentions (Column 5, Rows 4 and 5; Supplementary Material S10). These results offer partial evidence for H1c and H2c.

Moderators (H3, H4, H5)

Individuals' news media literacy did not moderate intervention effects; none of the moderation results were significant (H3 rejected). The level of misinformation concern generally did not moderate the intervention effects on risk perceptions and intentions, but it did moderate message credibility. There was no omnibus significant three-way interaction between intervention, accurate vs. inaccurate message, and misinformation concern ($F(2, 1092) = 2.45, p = 0.08$), yet the Tukey's post-hoc test revealed a significant pairwise difference: For those people with high pre-test misinformation concern and

Table 1. Effects of intervention (W1 manipulation factor: control vs. literacy training vs. interaction) on outcomes at W2

	Message credibility (W2)	Perceived risk (W2)	Perceived risk change (W2 – W1)	Intention (W2)	Intention change (W2 – W1)
All conditions <i>F</i> tests					
<i>Intervention</i>	$F(2, 1098) = 0.46$, ns	$F(2, 1098) = 0.48$, ns	$F(2, 1098) = 0.51$, ns	$F(2, 1098) = 3.05$, $p < .05$, $\eta_p^2 = 0.005$	$F(2, 1098) = 0.66$, ns
<i>Accurate vs. inaccurate</i>	$F(1, 1098) = 51.82$, $p < .001$, $\eta_p^2 = 0.05$	$F(1, 1098) = 2.45$, ns	$F(1, 1098) = 8.17$, $p < .01$, $\eta_p^2 = 0.007$	$F(1, 1098) = 7.04$, $p < .01$, $\eta_p^2 = 0.006$	$F(1, 1098) = 10.88$, $p < .001$, $\eta_p^2 = 0.009$
<i>Intervention × accurate vs. inaccurate</i>	$F(2, 1098) = 1.04$, ns	$F(2, 1098) = 4.81$, $p < .01$, $\eta_p^2 = .01$	$F(2, 1098) = 3.77$, $p < .05$, $\eta_p^2 = 0.007$	$F(2, 1098) = 0.99$, ns, $\eta_p^2 = 0.002$	$F(2, 1098) = 0.70$, ns, $\eta_p^2 = 0.001$
Post-hoc Tukey's tests	*	(1) Control seen incorrect – control seen correct, $M_{diff} = 0.09$, $p < .05$	(1) Control seen incorrect – control seen correct, $M_{diff} = 0.07$, $p < .05$	(1) Inoculation seen correct – control seen incorrect, $M_{diff} = 0.12$, $p < .01$ (2) Inoculation seen correct – literacy seen incorrect, $M_{diff} = 0.08$, $p < .10$	(1) Inoculation seen correct – control seen incorrect, $M_{diff} = 0.07$, $p < .10$ (2) Control seen correct – control seen incorrect, $M_{diff} = 0.08$, $p < 0.10$
Accurate info. conditions <i>F</i> tests					
<i>Intervention</i>	$F(2, 446) = 0.69$, ns	$F(2, 446) = 2.83$, $p < .10$, $\eta_p^2 = 0.009$	$F(2, 446) = 4.05$, $p < .05$, $\eta_p^2 = 0.02$	$F(2, 446) = 2.45$, $p < .10$, $\eta_p^2 = 0.01$	$F(2, 446) = 0.24$, ns.
Post-hoc Tukey's tests		Literacy – control, $M_{diff} = 0.07$, $p = .05$	Literacy – control, $M_{diff} = 0.06$, $p < .05$	Inoculation – literacy, $M_{diff} = 0.08$, $p < .10$	

Notes. Intention—getting the vaccine or not avoiding GMO consumption. C1 = Vaccine misinformation news; C2 = Vaccine misinformation social media post; C3 = Vaccine accurate info. news; C4 = Vaccine accurate info. social media post; C5 = GMO misinformation. news. W1 interventions: (Control vs. Lit. Training vs. Inoculation). Pairwise tests report on Tukey corrected p values. All conditions ($N = 1104$; C1 + C2 + C3 + C4 + C5) by accurate vs. inaccurate information (C3+C4 vs. C1+C2+C5); Vaccine accurate information conditions ($N = 449$; C3 + C4). Models that test all combinations are in [Supplementary Material S9](#). Models with additional controls produce either same or in many cases stronger differences (in both effect sizes and significance levels) ([Supplementary Material S10](#)).

* All nine pair-wise differences for credibility contrasting correct vs. incorrect are significant; but they are not about intervention effects.

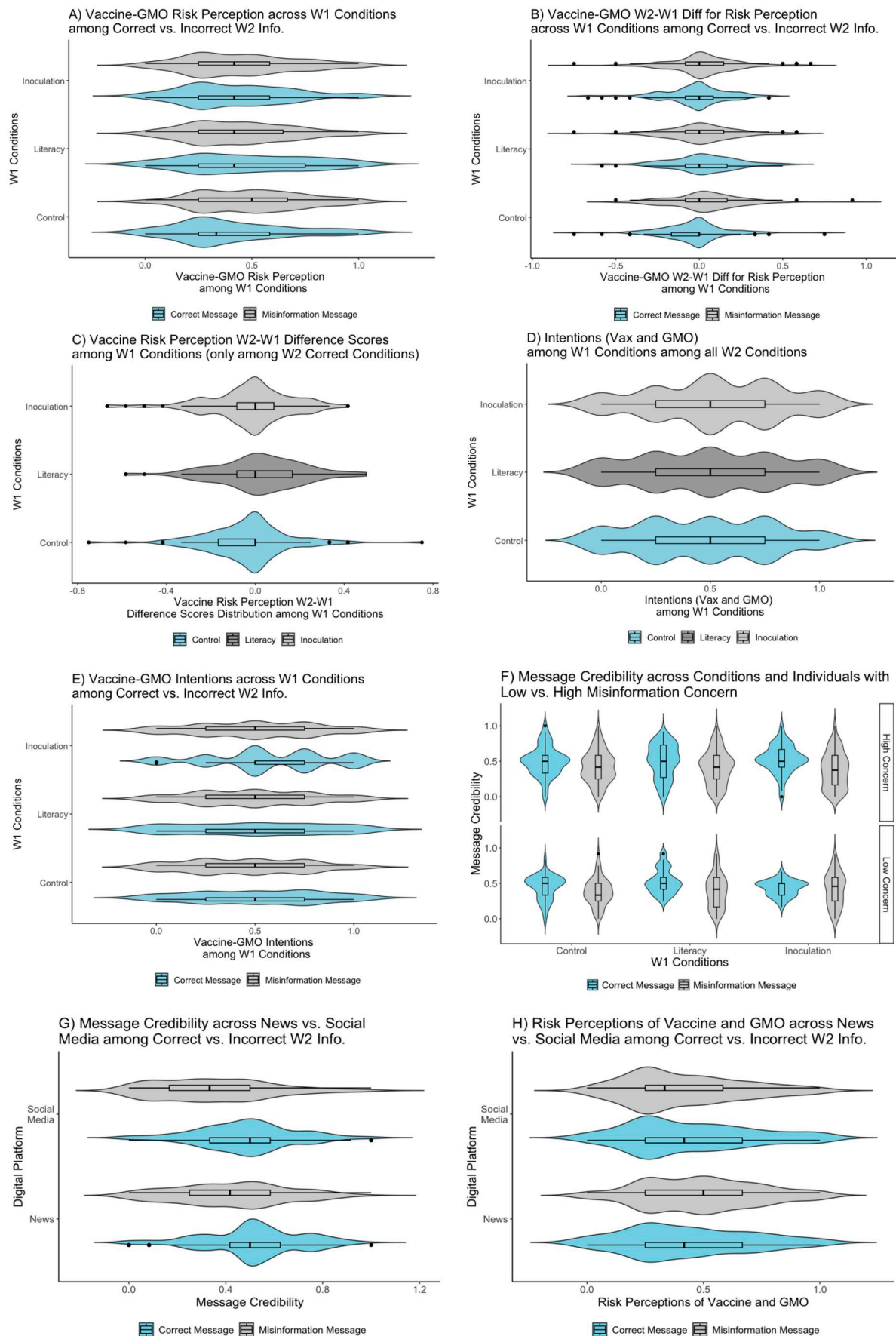


Figure 2. Visualizations of key effects across the different models and the outcomes variables.

received the inoculation, the difference between credibility of accurate and inaccurate messages in W2 was significant and largest (Tukey's test, $M_{\text{diff}} = 0.15$, $p < 0.001$). Those who had low levels of misinformation concern were indifferent in their credibility assessments of accurate and inaccurate messages. In Figure 2F, the difference between the top and bottom rows in the rightmost (inoculation) panel illustrated this effect (H4 partially supported).

We tested whether the message format moderated the interaction effects. None of the intervention interactions were significant. Hence, H5 is rejected. Yet, there were notable main effects. Among all conditions, social media posts were perceived as less credible than the messages delivered in the news story format, $F(1, 1102) = 7.66$, $p < 0.01$; Tukey's test; $M_{\text{diff}} = 0.04$, $p < 0.01$. This is illustrated in Figure 2G where social media conditions' message credibility (top two distributions) was lower than that of news (bottom two). This main effect, although not initially proposed as a separate hypothesis, is consistent with the idea that is inherently suggested in the logic of H5 (cf. Wasike, 2017). Moreover, all types of messages in social media format resulted in significantly lower risk perceptions overall (cross-sectionally), $F(1, 1102) = 6.33$, $p < 0.05$; Tukey's test; $M_{\text{diff}} = 0.04$, $p < .05$. Further, specifically, misinformation messages in the social media post format resulted in significantly lower risk perceptions than misinformation in the news format too (message accuracy—message format interaction, $F(1, 1102) = 6.17$, $p < 0.05$; Tukey's test, $M_{\text{diff}} = -0.07$, $p < 0.01$, Tukey's test). This is visible in Figure 2H. We conducted further (not preregistered) tests, which provided more context (Supplementary Material S10).

Further tests

Effect sizes ranged from $\eta_p^2 = 0.001$ to $\eta_p^2 = 0.05$ (Table 1), which could be considered mostly small ($\eta_p^2 > 0.005$). We also tested (a) demographic and other controls, (b) interference, (c) time variance, and (d) attrition, which provided additional context supportive of the main findings (Supplementary Material S10).

Discussion

This study compared the effectiveness of literacy training and inoculation interventions in the context of processing health statistics by tapping a very key, yet understudied, issue in health misinformation: the use of correlational data as evidence for causation. We investigated whether primarily information alone (literacy training) or the combination of information and motivational triggers (inoculation) is more effective in inducing a critical lens in processing statistics about COVID-19 vaccines and GMOs (cf. Jamieson et al., 2024; Jiang & Fang, 2019). The study provides a direct comparison of the two interventions and delineates the effects of the forewarning component while keeping constant the informational component. Further, three factors were investigated concerning their conditioning role on the relative effectiveness of the underlying informational and motivational components. The theory-driven comparison, the longitudinal design, the findings, and the insights offer important contributions and do so with evidence from a multicultural country in Southeast Asia that has four official languages (cf. Rad et al., 2018).

The first set of results illustrated the effectiveness of both interventions in reducing and preventing risk perceptions,

and to some extent, increasing behavioral intentions (for getting vaccinated and non-avoidance of GMOs). This finding shows that the effects of statistical misinformation on beliefs could be pre-emptively countered, at least to some extent. These results also offer the first set of evidence about the effectiveness of pre-emptive interventions against statistical misinformation, adding to the literature on post-hoc corrections (e.g., Irving et al., 2022; Vraga et al., 2019). The focus on statistical misinformation and the substantive nature of the training on “correlation does not imply causation” enrich the literature in crucial ways. Prior work on pseudo-science effects mostly examined the effects of descriptions of fictional viruses or measured “bullshit receptivity” to nonsense statements (Evans et al., 2020; O'Brien et al., 2021). While these studies provided important insights about pseudo-science, they did not engage with more scientific-looking realistic messages or the mis-contextualization of accurate statistics, as commonly observed in I-do-my-own-research movements (Chinn & Hasell, 2023). Instead, the current study focused on more nuanced and misleading types of (mis)information. We found that anti-vaccine discourse misrepresenting correlational data (Jamieson et al., 2024) or unwarranted causal claims about the consequences of consuming GMOs (Jiang & Fang, 2019) could be countered through methodological and statistical training (cf. Peters, 2020).

Second, we found evidence that inoculation was slightly more effective than literacy training, both in the extent of the effects and in avoiding spill-over (cynicism) effects (Guess et al., 2020; Hameleers, 2023). This finding is important as it offers one plausible explanation for why literacy interventions may result in backfiring while processing accurate information. An explicit motivational trigger against the threat, as in inoculation, may help individuals specify, target, and narrow down their critical lens only on questionable contents (e.g., hyped causality claims) instead of applying it indiscriminately to all statistical evidence about health and science. This finding also speaks to the importance of understanding the complexity of threat in inoculation (Amazeen et al., 2022; Compton, 2021). However, other work shows that *misinformation warnings alone* may also result in cynicism (van der Meer et al., 2023). This suggests that the best approach is not (solely or predominantly) informational intervention and not (solely or predominantly) motivational intervention, but rather their (more balanced and integrated) combination, a position for which our results provide support. Future work should directly compare all three approaches simultaneously (informational, motivational, or the balanced/mixed approach) to settle this insight confidently.

Third, our results contribute to the communication intervention science, particularly in comparing competing pre-emptive approaches, which are currently drawing extensive attention. Most studies comparing different communication interventions to date have adopted “the toolbox approach” as they included numerous techniques, to the extent possible, to be comprehensive in a maturing field where practical solutions that move the needle are desperately needed (Arechar et al., 2023; Fazio et al., 2024; Hoes et al., 2024; Kozyreva et al., 2023; Spampatti et al., 2024). While these studies provide important comparative evidence and recommendations, we also need more theory-driven comparisons to complement these efforts (cf. Amazeen & Krishna, 2024). The current study is one of the first attempts to answer this call. We compared two distinct interventions, motivated by theory, and thus brought empirical evidence to the debates surrounding

literacy training and inoculation, particularly the latter's distinction from the former. We argued and found evidence for inoculation's somewhat marginal advantage as it delineates, more clearly, a separate and explicit motivational trigger that may guide individuals with both mental resource mobilization and selective targeting in applying their critical lens. This theory-driven comparative framework can be implemented in explicating and comparing other interventions as well.

The current study also speaks to the theoretical debates in communication intervention science. We show that it is crucial to interrogate how the underlying fundamental components (information and motivation) of both approaches (literacy training and inoculation) can operate independently. The current design makes this comparison directly by distinguishing between more surface-level vs. underlying differences. We suspect that the overlooked similarities between literacy and inoculation are, at least partly, the manifestation of the theoretical competition—the boundary work—in the field. This is because many well-known communication phenomena (e.g., rumors, misunderstandings) get reconceptualized as “infodemic” and “inoculation against misinformation virus” in recent research (Williams, 2023). As the field moves towards epistemological pluralism (Tenenboim-Weinblatt & Lee, 2020), it is crucial to cultivate theoretical debates by being more conscious about the storytelling distinctions in social science theories (i.e., vaccine analogy) (Gelman & Basbøll, 2014). We conclude this section by pointing out that it is possible to incorporate motivational triggers into literacy interventions without theorizing or labeling them as “inoculation.” We agree that misinformation and interventions against it could be less deterministically (e.g., DNA, virus, inoculation) characterized (Freiling et al., 2023; Williams, 2023). This point also speaks to the critiques of highly influential inoculation interventions on the ethical front. The interventions' deterministic and sometimes simplistic emphasis on characterizing misinformation as the source instead of the symptom of the larger structural, political, and social issues is problematic (Freiling et al., 2023; Williams, 2023). Our findings about the threat (of misinformation) show the important role of framing threats in interventions.

Fourth, we offered multi-platform evidence by comparing information presented in the news vs. social media format and investigated how individual differences in news literacy behaviors and misinformation concern levels may matter in reacting to the underlying informational and motivational components. We found that social media content is perceived as less credible, leading to lower risk perceptions. This makes sense because people would discount the evidence on social media more (cf. Metzger & Flanagin, 2013). These results replicate other findings in the literature on platform effects (Wasike, 2017). Yet, the format of the messages about vaccines and GMOs that individuals encountered did not moderate the intervention effects. We also did not find any moderation effects driven by individuals' preexisting levels of news literacy behaviors. However, we found that individuals with high misinformation concern levels reacted more strongly to inoculation (cf. Altay & Acerbi, 2023). The effects of forewarning on message credibility were contingent on pre-existing misinformation concern levels. In other words, inoculation reduced the credibility of misinformation only for individuals with high pre-existing misinformation concern levels. Offering the first evidence about motivational

individual differences in responsivity to forewarning (cf. Compton, 2021), this result gives additional support to the value of forewarning and calls for more work on the conditioning role of such individual factors to understand the nature of heterogeneous treatment effects.

Methodologically, the two-wave data collection and delayed outcome measurement strategy offered a conservative testing of interventions with stronger ecological validity. Most research in this domain relied on pre- and post-manipulation measurement of outcomes within cross-sectional studies, giving rise to concerns about demand effects (cf. Chinn & Hart, 2021). Our study is one of the few to measure the outcomes at a delayed time instead of immediately following the interventions (Maertens et al., 2023). Indeed, misinformation interventions decay rapidly without an immediate post-test (Capewell et al., 2023). The fact that we found generally small but significant effects, despite the gap between the two waves (median = 21 days), shows the promise of substantive and focused training. On the practical front, we showed the importance of cautioning the public to employ their newly learned and/or refreshed statistical skills and methodological knowledge in a more targeted manner. While giving examples, practitioners can consider whether to induce forewarning either directly or indirectly by clarifying how that example may invoke threat perceptions. Implementation should also be mindful of different dimensions of threat in specific applied contexts (Ivanov, 2017) and transparency concerning ethical issues (Williams, 2023).

Limitations and Future research

There are a few limitations and important questions to be addressed in future research. First, the evidence from a single country limits the applicability to other contexts; Singapore has relatively less politicization of science and high trust in health authorities. Also, larger N studies are needed to get more confident results on how the effects of interventions might decay over time. Similarly, some three-way interactions are slightly underpowered; especially, the null finding on the moderating role of news literacy behaviors should be taken cautiously. As with any longitudinal design, individuals returning to take the second survey differed demographically. The pre-registration of hypotheses and analyses included unclear parts or needed more comparison details; more robust preregistered analysis plans will ensure greater precision in future work. It is also important to test these interventions in more substantive and less controlled settings via field experiments (Badrinathan, 2021). Future research can also investigate the potential effects of different informational components (e.g., media literacy, digital news literacy) and motivational triggers (e.g., in health vs. political contexts). Finally, future interventions should integrate insights from health inequalities (Freimuth & Quinn, 2004) and critical studies more centrally (Ramasubramanian & Banjo, 2020). Training and motivating people to critically process health statistics is only one (individual level) solution, among the numerous other social and structural ones, in containing the multi-faceted challenges of digital health communication at large.

Supplementary material

Supplementary material is available online at *Journal of Communication* online.

Data availability

The data and code for this study are deposited at OSF: <https://osf.io/gmwra/?view_only=188dc3411fba4f1db6676600db0f67bd>

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Pre-registration

The experimental protocol has been preregistered at AsPredicted.org before the data collection started: #143007. <https://aspredicted.org/ZG4_57Q>.

Ethics

This study has received IRB clearance at the Institutional Review Board of the National University of Singapore: NUS-IRB-2023-555.

Diversity, labor, and generative AI disclosures

The author carefully checked the research referenced and the language used for diversity and inclusion. There is no unreported/unacknowledged research assistance. Generative artificial intelligence, such as ChatGPT, is not used at any stage of this study.

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